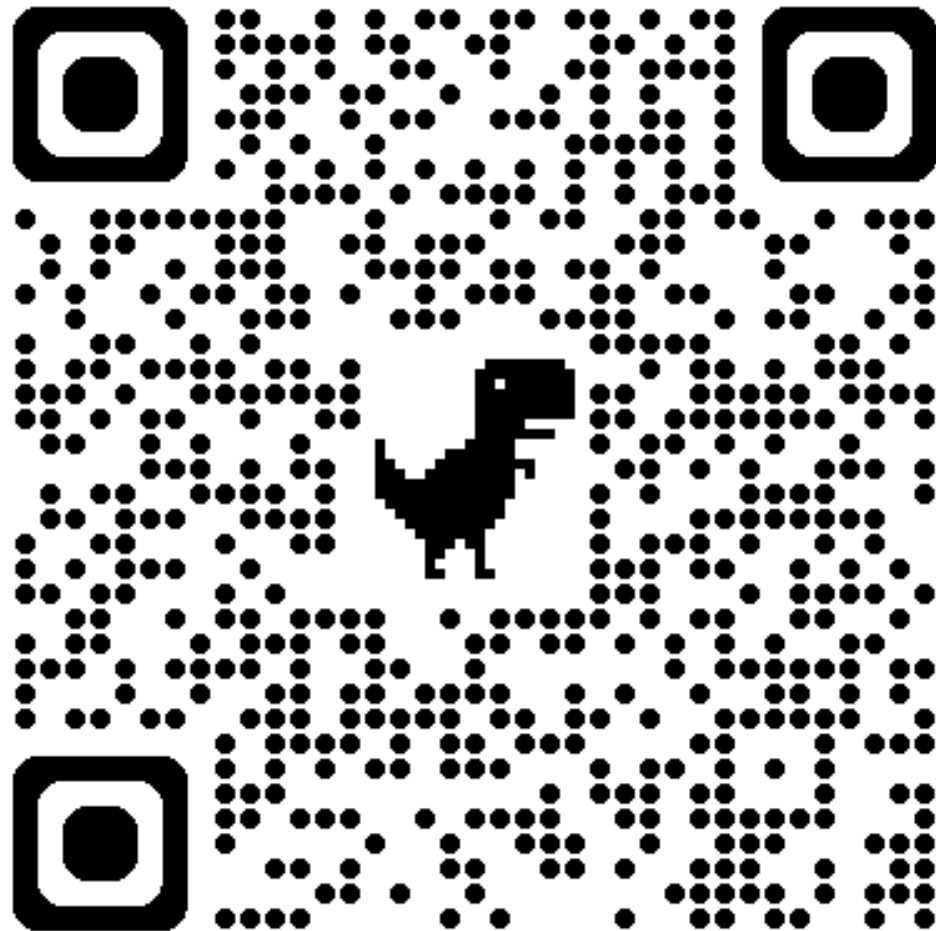


Course
Website

114-2

自然語言處理與
應用



https://github.com/mcps5601/CGUNLP_2026_Spring



自然語言處理與應用

Natural Language Processing and Applications

Course Introduction

Instructor: 林英嘉 (Ying-Jia Lin)
2026/02/23



Slido
NLP0223 ([Link](#))

About Me

Experience and Education (Time)		
Assistant Professor	Department of Artificial Intelligence, Chang Gung University (2025/02 -)	
Postdoctoral Researcher	Department of Computer Science, National Tsing Hua University (2024/09 - 2025/01)	
Ph.D.	Department of Computer Science and Information Engineering, National Cheng Kung University (2019/08 - 2024/02)	
M.S.	Institute of Biomedical Informatics, National Yang-Ming University (2017/09 - 2019/07)	
B.S.	Department of Biomedical Sciences, Chang Gung University (2013/09 - 2017/06)	



Outline

- Introduction to Natural Language Processing
- Introduction to Large Language Models
- Course Introduction and requirements

什麼是自然語言處理 (Natural Language Processing, NLP) ?

文字搜尋

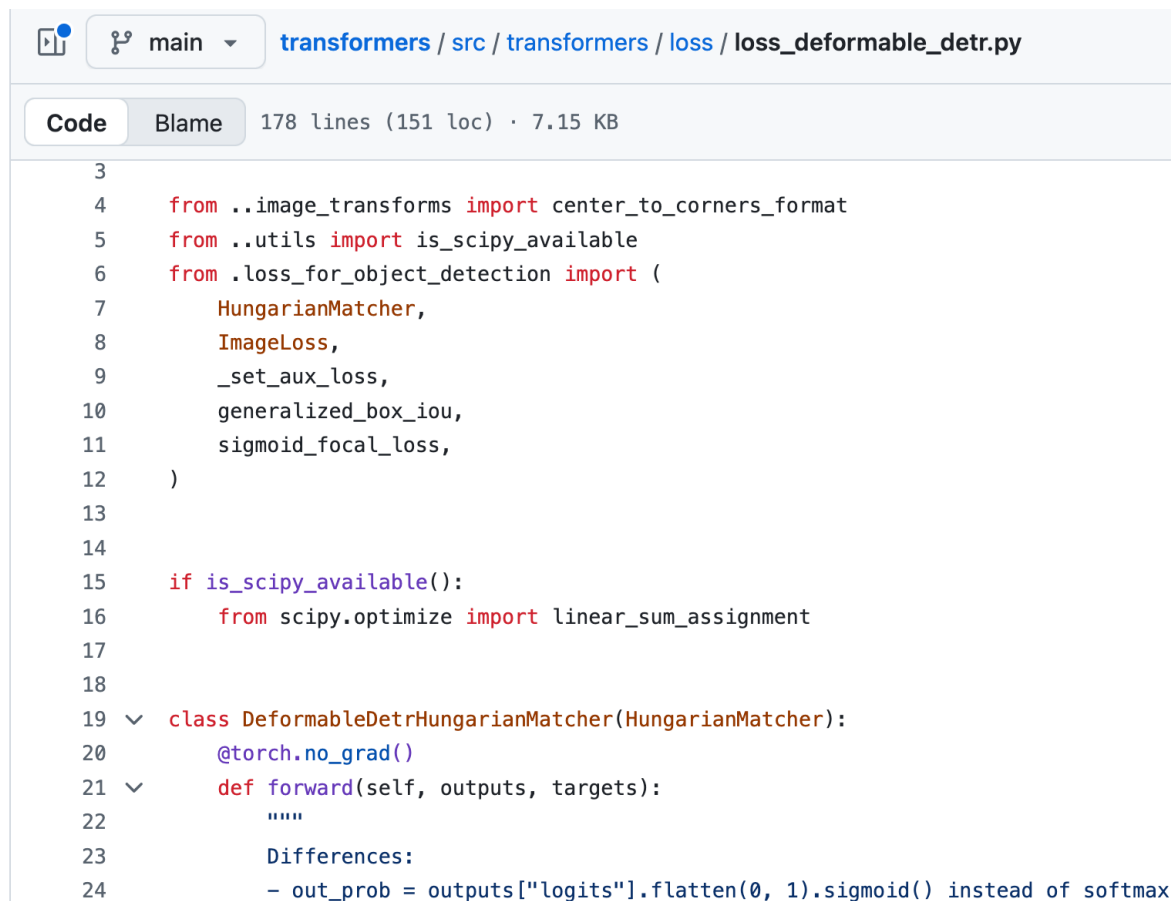


下一個字預測



程式語言 vs. 自然語言

程式語言



The screenshot shows a code editor interface with a file path: `transformers / src / transformers / loss / loss_deformable_detr.py`. The code is as follows:

```
3
4 from ..image_transforms import center_to_corners_format
5 from ..utils import is_scipy_available
6 from .loss_for_object_detection import (
7     HungarianMatcher,
8     ImageLoss,
9     _set_aux_loss,
10    generalized_box_iou,
11    sigmoid_focal_loss,
12 )
13
14
15 if is_scipy_available():
16     from scipy.optimize import linear_sum_assignment
17
18
19 class DeformableDetrHungarianMatcher(HungarianMatcher):
20     @torch.no_grad()
21     def forward(self, outputs, targets):
22         """
23         Differences:
24         - out_prob = outputs["logits"].flatten(0, 1).sigmoid() instead of softmax
```

自然語言

先生不知何許人也，亦不詳其姓字。宅邊有五柳樹，因以為號焉

閑靜少言，不慕榮利。好讀書，不求甚解，每有會意，便欣然忘食。性嗜酒，家貧，不能常得。親舊知其如此，或置酒而招之。造飲輒盡，期在必醉，既醉而退，曾不吝情去留。環堵蕭然，不蔽風日；短褐穿結，簞瓢屢空。——晏如也。常著文章自娛，頗示己志。忘懷得失，以此自終。

贊曰：黔婁之妻有言：「不戚戚於貧賤，不汲汲於富貴。」極其言，茲若人儔乎？酣觴賦詩，以樂其志。無懷氏之民歟！葛天氏之民歟！

什麼是自然語言處理？



定義 ➡

自然語言處理（ Natural Language Processing, NLP ）
是讓電腦能「理解、分析、產生」人類語言的技術。

The Revolution of ChatGPT

ChatGPT came out in November, 2022.

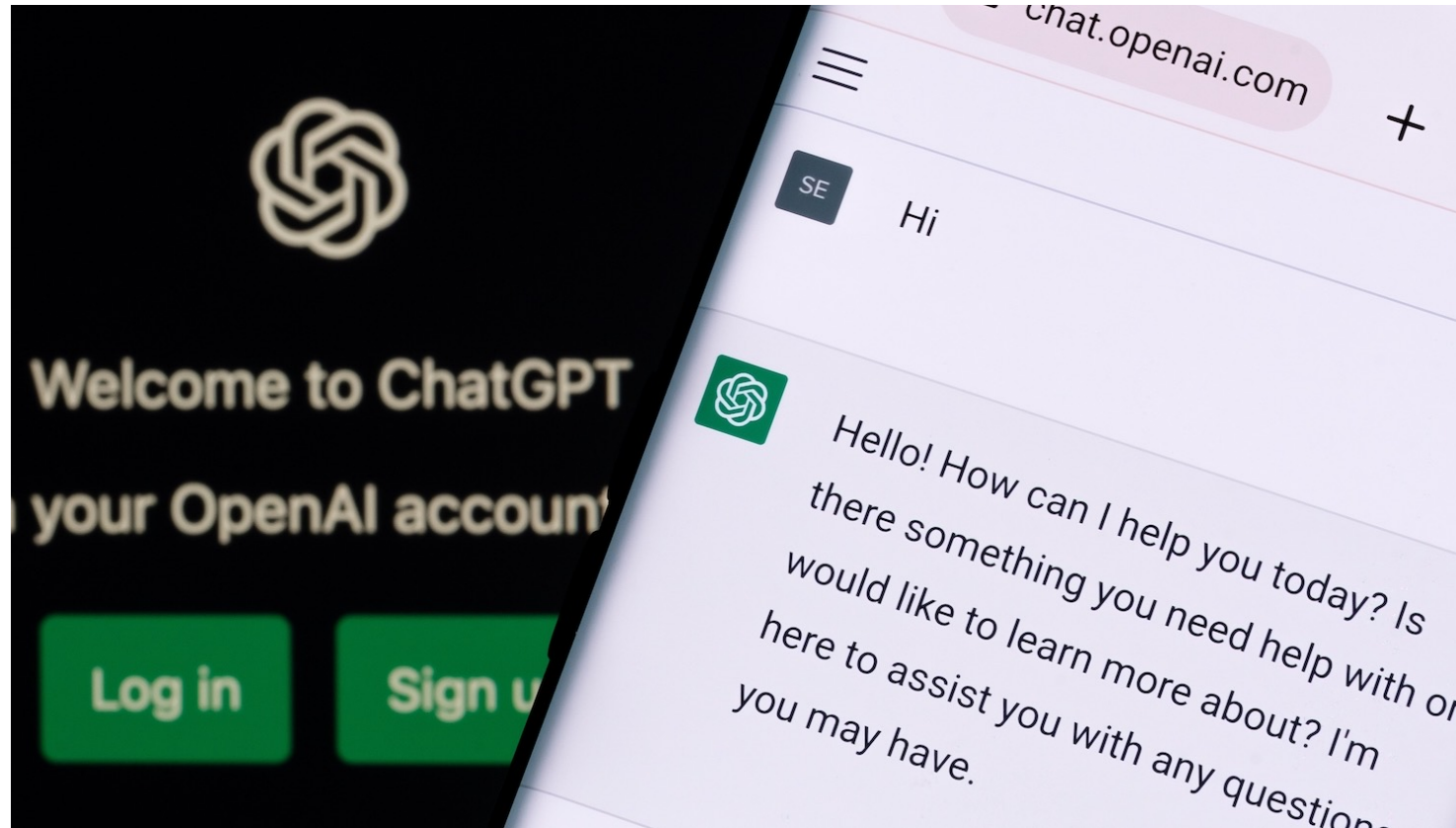
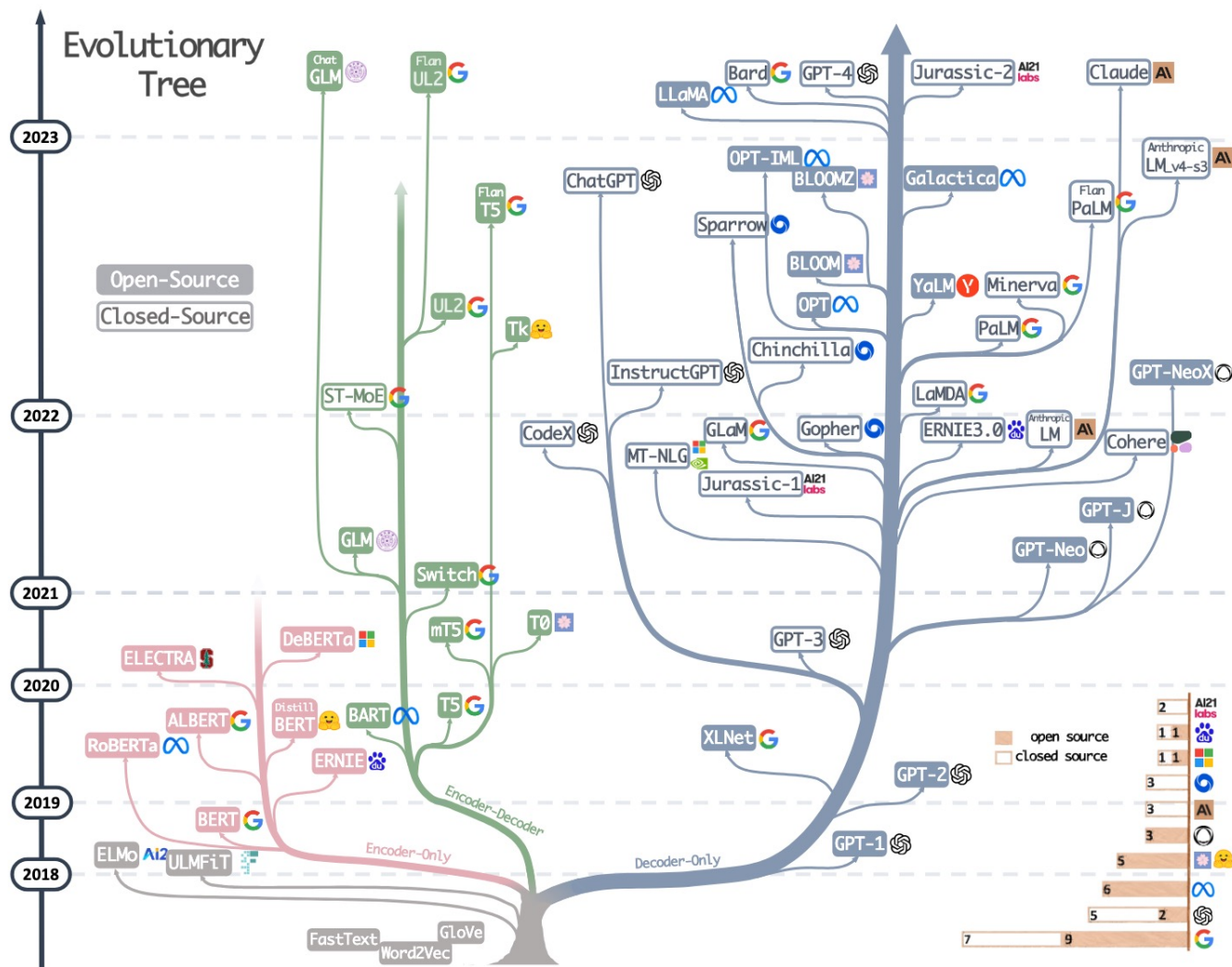


Figure source: <https://poole.ncsu.edu/thought-leadership/article/lets-chat-about-chatgpt/>
<https://openai.com/index/chatgpt/>

大型語言模型的演化樹



Yang, Jingfeng, et al. "Harnessing the power of llms in practice: A survey on chatgpt and beyond." ACM Transactions on Knowledge Discovery from Data 18.6 (2024): 1-32.

Chinese LLMs

SuperCLUE: 2024年值得关注的中文大模型全景图



SuperCLUE Leaderboard

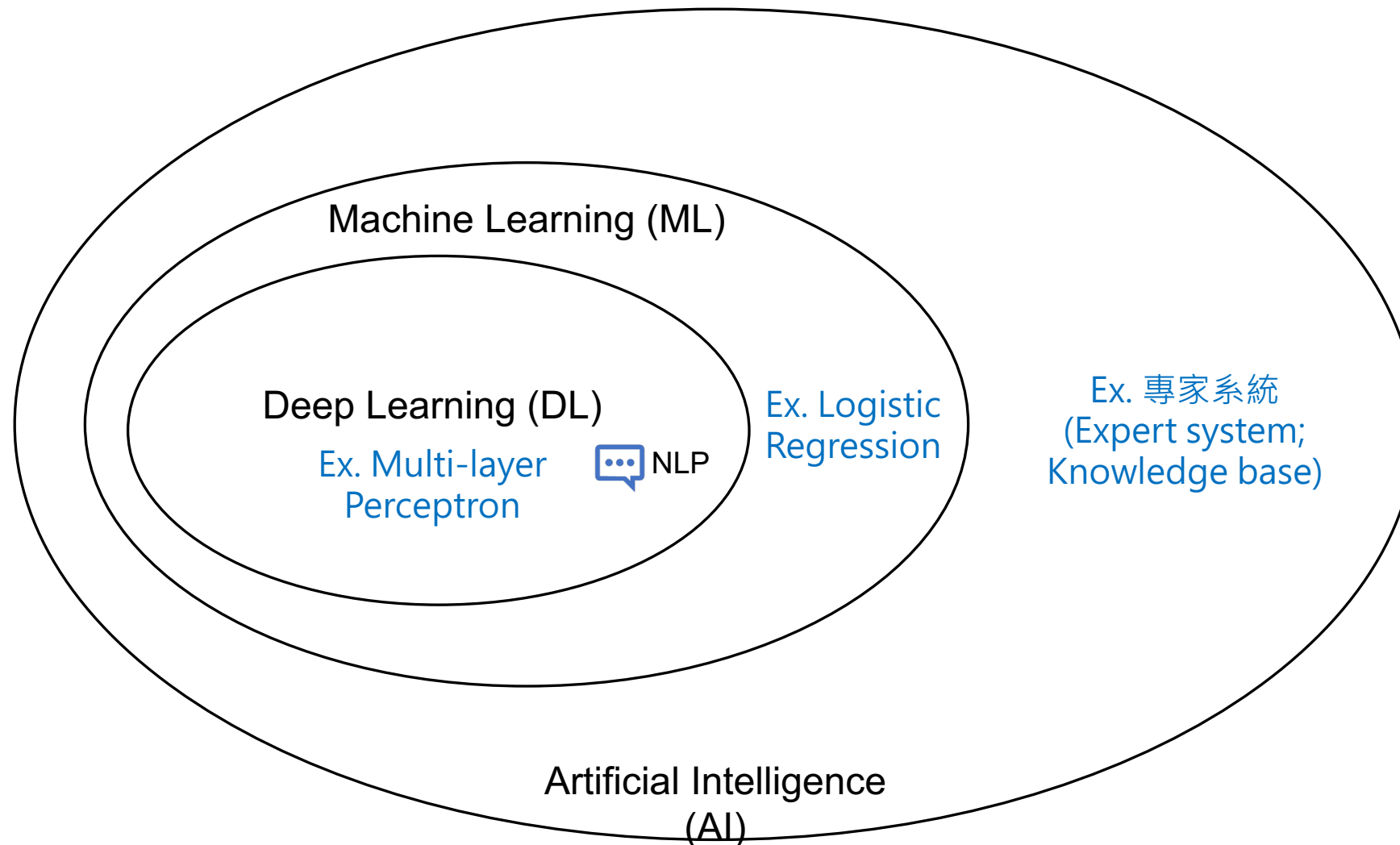
为减少波动影响，榜单将分差1分值内的模型视为并列排名。国外模型与部分国内模型不参与排名，只做参考。

排名	模型名称	机构	开/闭源	总分	数学推理	幻觉控制	科学推理	精确指令遵循	代码生成	智能体(任务规划)	属地	使用方式	是否推理	发布日期
-	Claude-Opus-4.6(high effort)	Anthropic	闭源	71.93	77.88	85.87	78.63	51.78	59.91	77.52	海外	API	是	2026.2.11
-	Claude-Opus-4.5-Reasoning	Anthropic	闭源	68.25	73.04	88.31	73.77	51.10	48.40	74.87	海外	API	是	2026.1.28
-	Gemini-3-Pro-Preview	Google	闭源	65.59	80.87	83.16	73.77	43.56	47.17	65.02	海外	API	是	2026.1.28
-	GPT-5.2(high)	OpenAI	闭源	64.32	72.04	88.56	75.21	37.81	30.91	81.39	海外	API	是	2026.1.28
1	Kimi-K2.5-Thinking	月之暗面	开源	61.50	77.39	78.54	67.21	24.45	53.33	68.06	国内	API	是	2026.1.29
-	Gemini-3-Flash-Preview	Google	闭源	61.43	79.13	81.57	74.17	39.45	40.64	53.59	海外	API	是	2026.1.28
1	Qwen3-Max-Thinking	阿里巴巴	闭源	60.61	80.87	74.05	68.85	28.22	41.56	70.13	国内	API	是	2026.1.29
1	GLM-5	智谱AI	开源	60.50	73.91	81.05	67.21	27.35	51.35	62.12	国内	API	是	2026.2.12
2	Doubao-Seed-1.8-251228(Thinking)	字节跳动	闭源	58.17	68.70	80.37	68.85	32.60	40.33	58.15	国内	API	是	2026.1.28
2	DeepSeek-V3.2-Thinking	深度求索	开源	57.55	72.17	76.57	71.31	29.86	39.84	55.53	国内	API	是	2026.1.28
3	GLM-4.7	智谱AI	开源	56.22	68.42	83.85	66.39	21.37	41.26	56.03	国内	API	是	2026.1.28
3	ERNIE-5.0	百度	闭源	56.18	65.22	74.61	64.75	37.53	45.63	49.32	国内	API	是	2026.1.28
-	Grok-4	X.AI	闭源	55.23	74.78	78.90	68.03	14.79	49.51	45.36	海外	API	是	2026.1.28
4	Kimi-K2-Thinking	月之暗面	开源	54.02	70.43	69.39	62.30	16.99	46.00	59.01	国内	API	是	2026.1.28
5	Qwen3-Max-Preview-Thinking	阿里巴巴	闭源	52.46	61.74	62.01	63.93	24.66	37.99	64.40	国内	API	是	2026.1.28
-	Grok-4.1-Fast(Reasoning)	X.AI	闭源	51.59	71.30	71.08	62.30	16.44	34.36	54.05	海外	API	是	2026.1.28
6	Tencent HY 2.0 Think	腾讯	闭源	50.22	65.79	70.91	66.39	18.90	37.32	41.99	国内	API	是	2026.1.29
6	LongCat-Flash-Thinking-2601	美团	开源	49.22	62.62	66.76	60.66	15.07	37.81	52.43	国内	API	是	2026.1.28
7	Step-3.5-Flash	阶跃星辰	开源	48.97	75.65	57.04	63.93	10.68	36.64	49.88	国内	API	是	2026.2.5
7	Qwen3-Max-2025-09-23	阿里巴巴	闭源	48.46	62.61	64.58	61.48	6.30	47.23	48.59	国内	API	否	2026.1.28

Figure source: <https://www.superclueai.com/>

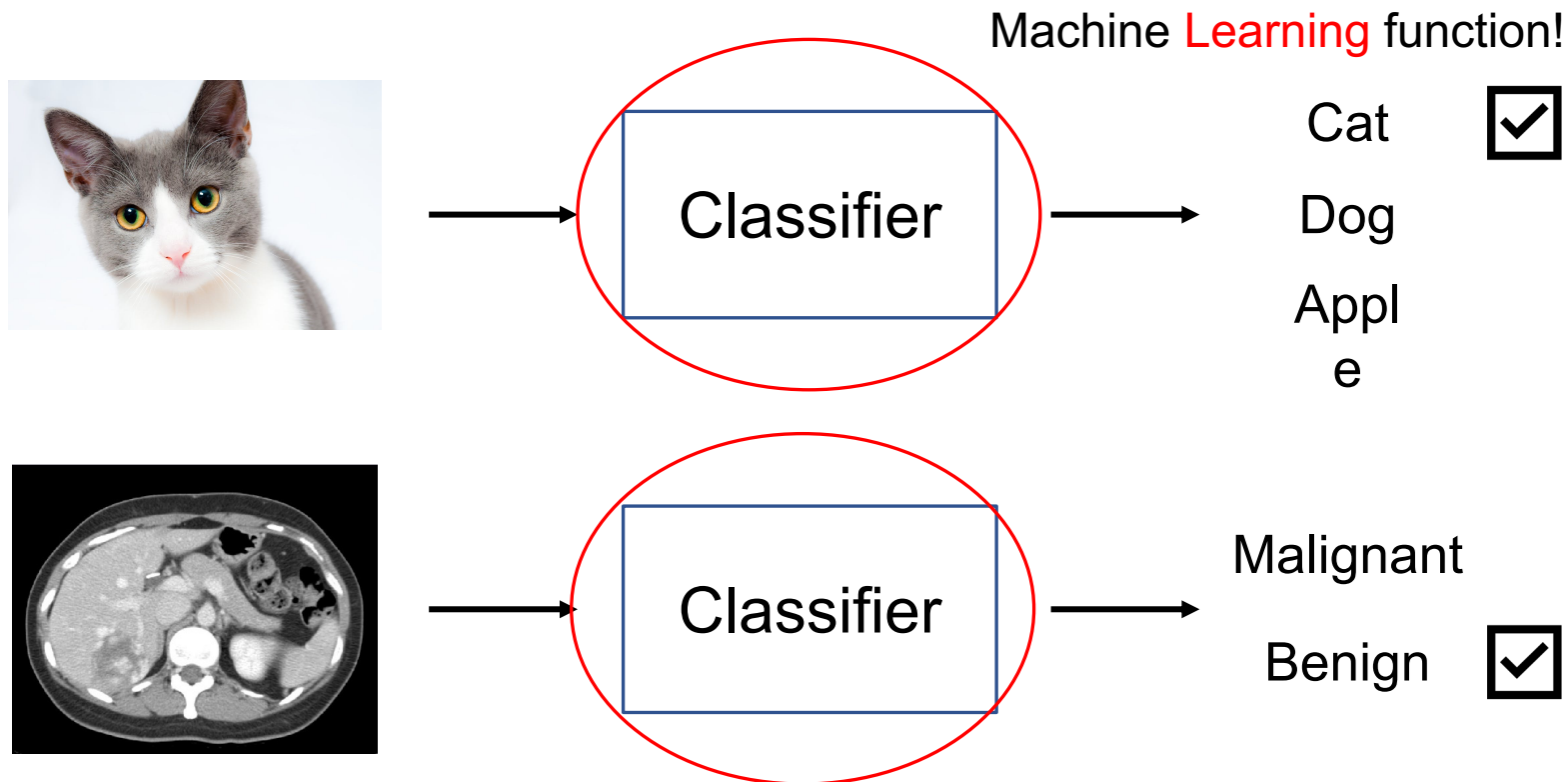
自然語言處理（廣義）屬於一
機器學習領域的任務

Venn Diagram about AI, ML, and DL



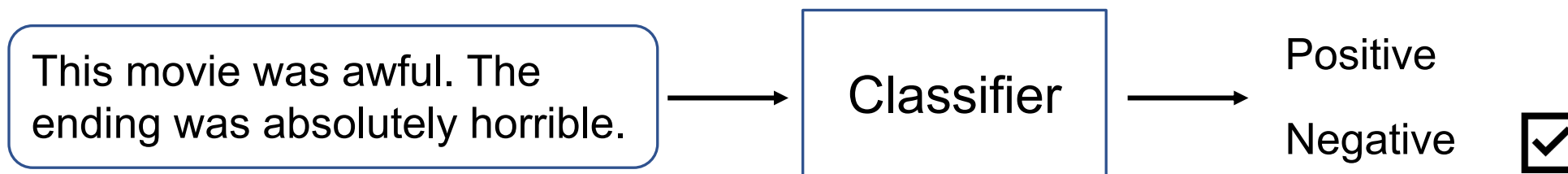
What is Deep Learning?

- We start the concepts from machine learning.

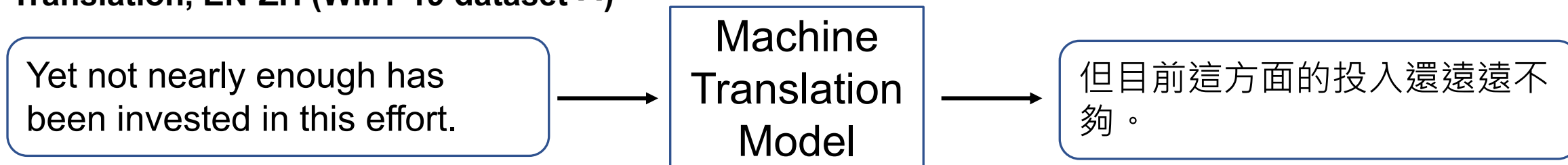


How about DL in Natural Language Processing?

Sentiment Analysis (IMDb dataset ^[1])



Translation, EN-ZH (WMT-19 dataset ^[2])



[1] <https://huggingface.co/datasets/stanfordnlp/imdb>

[2] <https://huggingface.co/datasets/wmt/wmt19/viewer/zh-en>

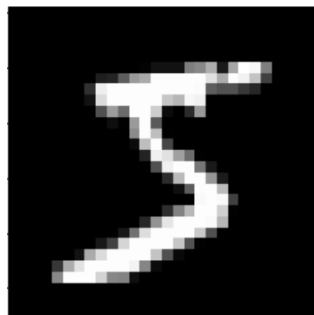
Why do we need DL in NLP?

79	RNTN	85.4	×	Recursive Deep Models for Semantic Compositionality Over a Sentiment Treebank			2013
80	SWEM-concat	84.3	×	Baseline Needs More Love: On Simple Word-Embedding-Based Models and Associated Pooling Mechanisms			2018
81	MV-RNN	82.9	×	Recursive Deep Models for Semantic Compositionality Over a Sentiment Treebank			2013
82	GloVe+Emo2Vec	82.3	×	Emo2Vec: Learning Generalized Emotion Representation by Multi-task Training			2018
83	Emo2Vec	81.2	×	Emo2Vec: Learning Generalized Emotion Representation by Multi-task Training			2018
84	ToWE-CBOW	78.8	×	Task-oriented Word Embedding for Text Classification			2018
85	Joined Model Multi-tasking	54.72	×	Exploring Joint Neural Model for Sentence Level Discourse Parsing and Sentiment Analysis			2017

<https://paperswithcode.com/sota/sentiment-analysis-on-sst-2-binary>
Liu, Qian, et al. "Task-oriented word embedding for text classification." Proceedings of the 27th international conference on computational linguistics. 2018.

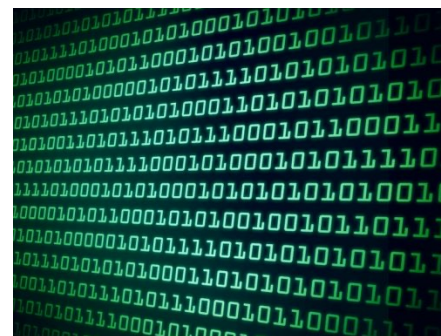
電腦是如何學習自然語言呢？

影像 vs. 文字



→ 28 x 28 的矩陣

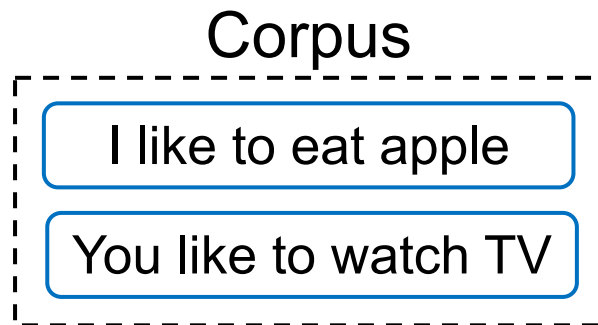
I want to study AI → ?



Word Representations



Basic Approach



→ Vocabulary

apple	[1, 0, 0, 0, 0, 0, 0, 0]
eat	[0, 1, 0, 0, 0, 0, 0, 0]
I	[0, 0, 1, 0, 0, 0, 0, 0]
like	[0, 0, 0, 1, 0, 0, 0, 0]
watch	[0, 0, 0, 0, 1, 0, 0, 0]
to	[0, 0, 0, 0, 0, 1, 0, 0]
⋮	

One hot encodings with each vector size equal to the vocab size (8 in this case)

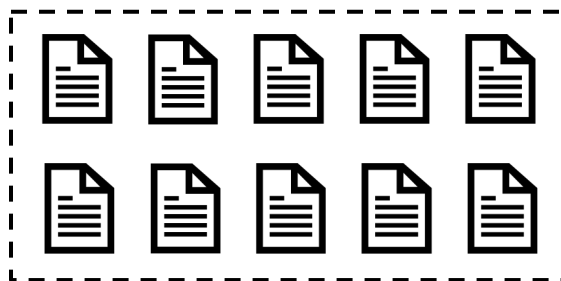
Pros: handy, training-free

Cons: sparse matrix which occupies much memory

詞嵌入：把詞嵌入到向量 (Word2Vec)

[1] Mikolov et al. "Distributed representations of words and phrases and their compositionality." NeurIPS 2013.

非常大量的語料庫
(例如：Wikipedia)



Input

Steve
Jobs
[MASK]
Apple
Inc.



CBOW ^[1]
(Continuous
Bag-of-Words)



Prediction

founded

[MASK]
[MASK]
founde
d
[MASK]
[MASK]



Skip-gram ^[1]



Steve
Jobs

Apple
Inc.

每個字，都是一段向量 – 詞向量

- Word2Vec 的範例長相:

Dimension size (e.g., 300)	
Vocabulary size (e.g., 30,000)	apple -0.110960 0.016115 -0.004809 0.033589 0.121455 ...
	banana -0.027713 -0.015676 0.003314 0.077602 0.159718 ...
	...
	...

Word Embeddings 視覺化



- **Word embedding model:** glove-wiki-gigaword-100
- **Dimension reduction:** t-SNE
- **Dataset:** Mikolov et al., 2013

Problems of Traditional Word Embeddings

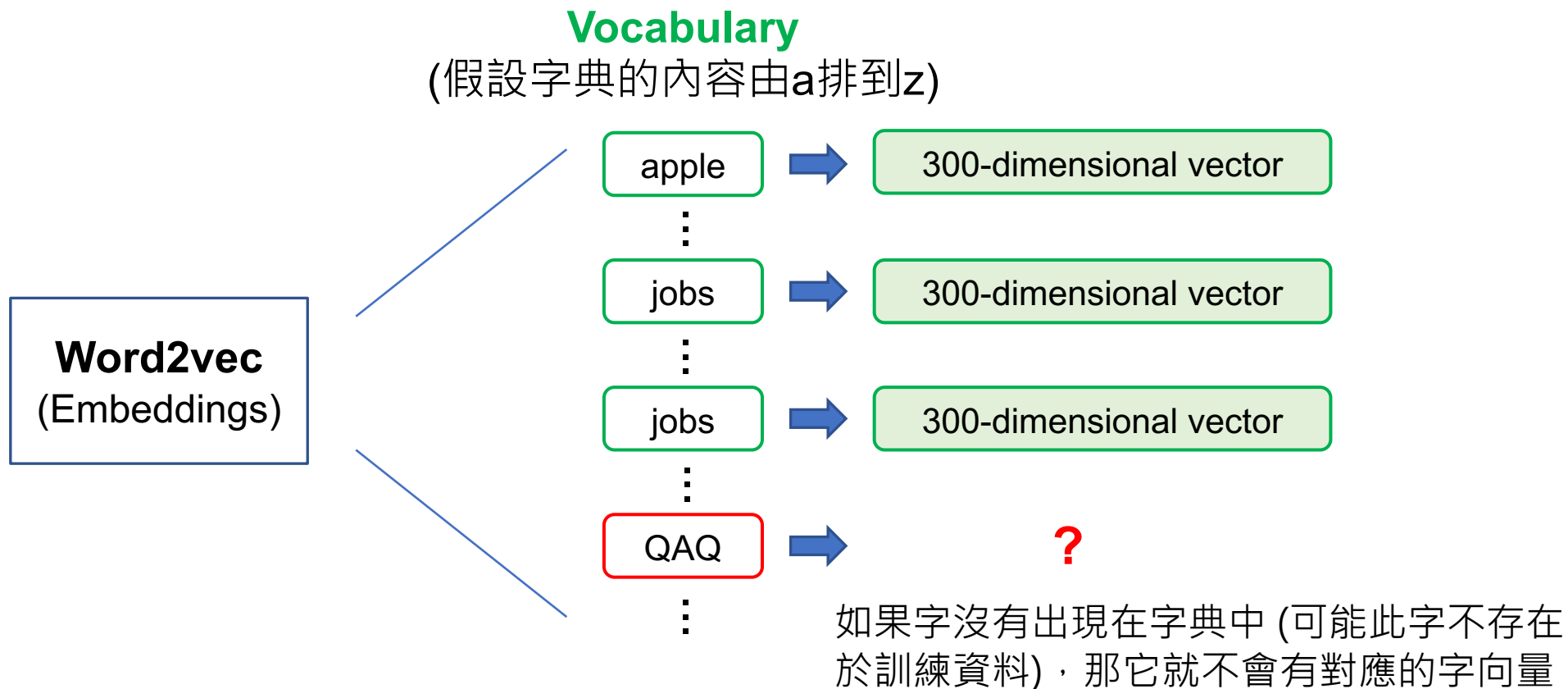
1. Out-of-vocabulary problem

2. Model architecture for down-stream applications

- Context-dependent representations

The Out-of-vocabulary Problem

- Before training word embeddings, words are split by whitespaces for English corpora



Sub-word Tokenization (Byte Pair Encoding)

- Byte Pair Encoding (BPE)^[3] is a **sub-word** (子詞) tokenization technique used to handle rare words by breaking them into more frequent sub-word units.

Example sentence:

I printed Hello world.

Traditional word tokenization

I printed Hello world

Traditional with **BPE**

I prin ted Hell o world

分詞方法的決定是無監督式的

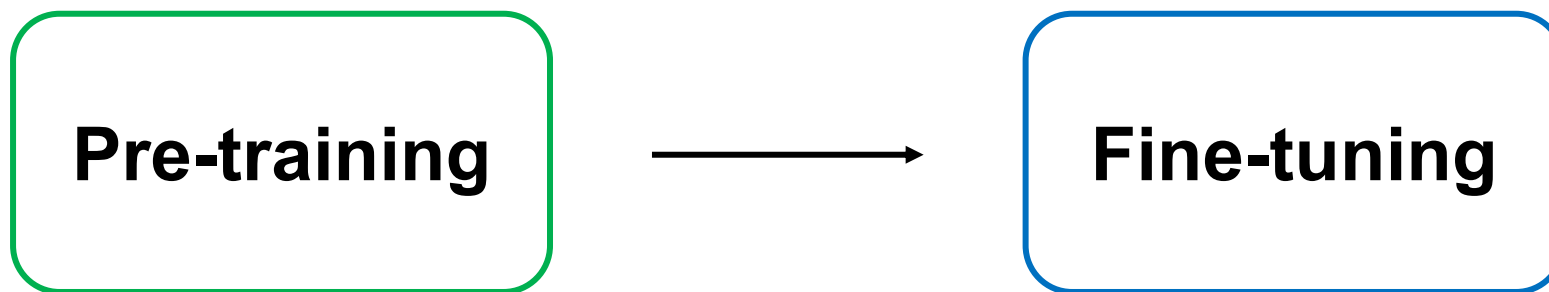
Sub-word Tokenization 的好處

- With sub-word tokenization algorithms like BPE, we can handle representations for **unknown words** (or **mis-spelled** words).
- In machine translation , the compound word issues between source and target languages can be alleviated.
- State-of-the-art pre-trained language models (e.g., GPT-3, BERT) adopt sub-word tokenization algorithms before pre-training.

Problems of Traditional Word Embeddings

1. Out-of-vocabulary problem
2. Model architecture for down-stream applications
 - Context-dependent representations

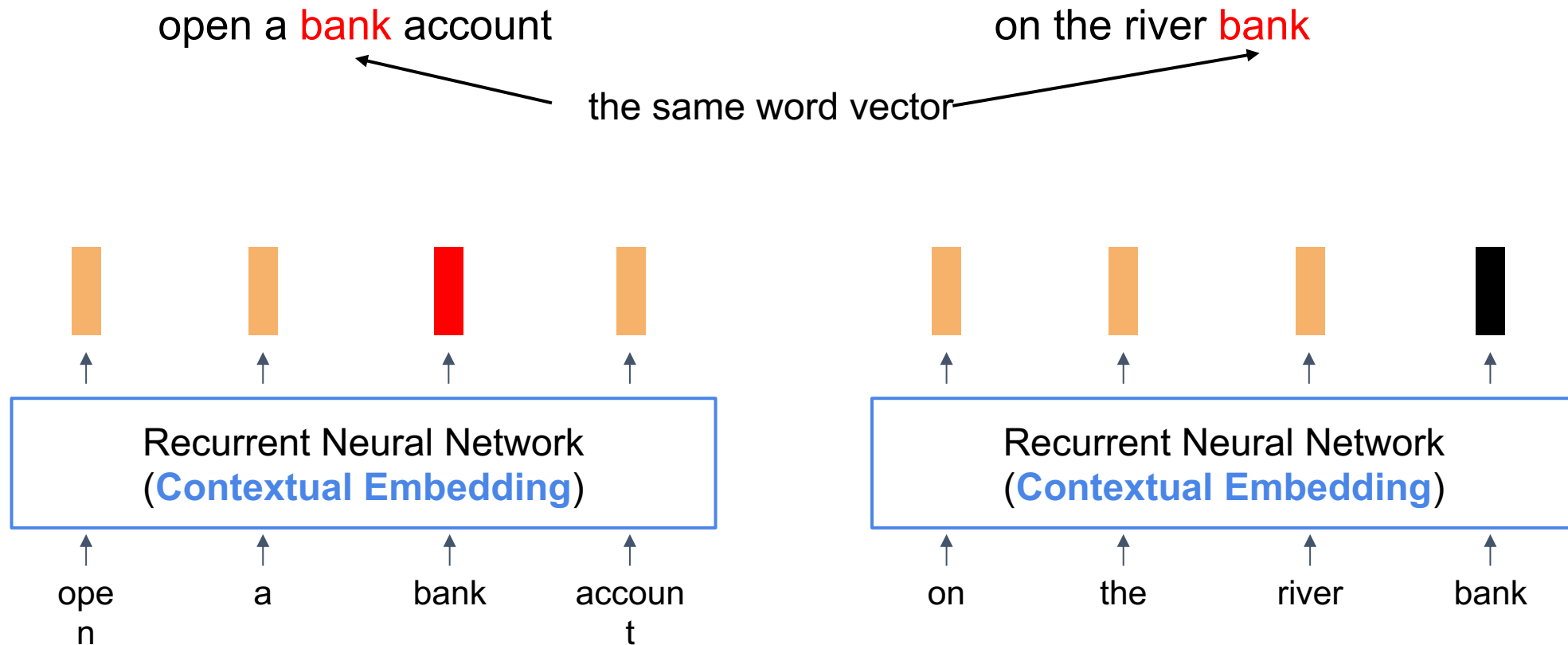
The Pre-training then Fine-tuning Paradigm



在大量資料上進行訓練，通常是自監督式 (Self-Supervised Training)

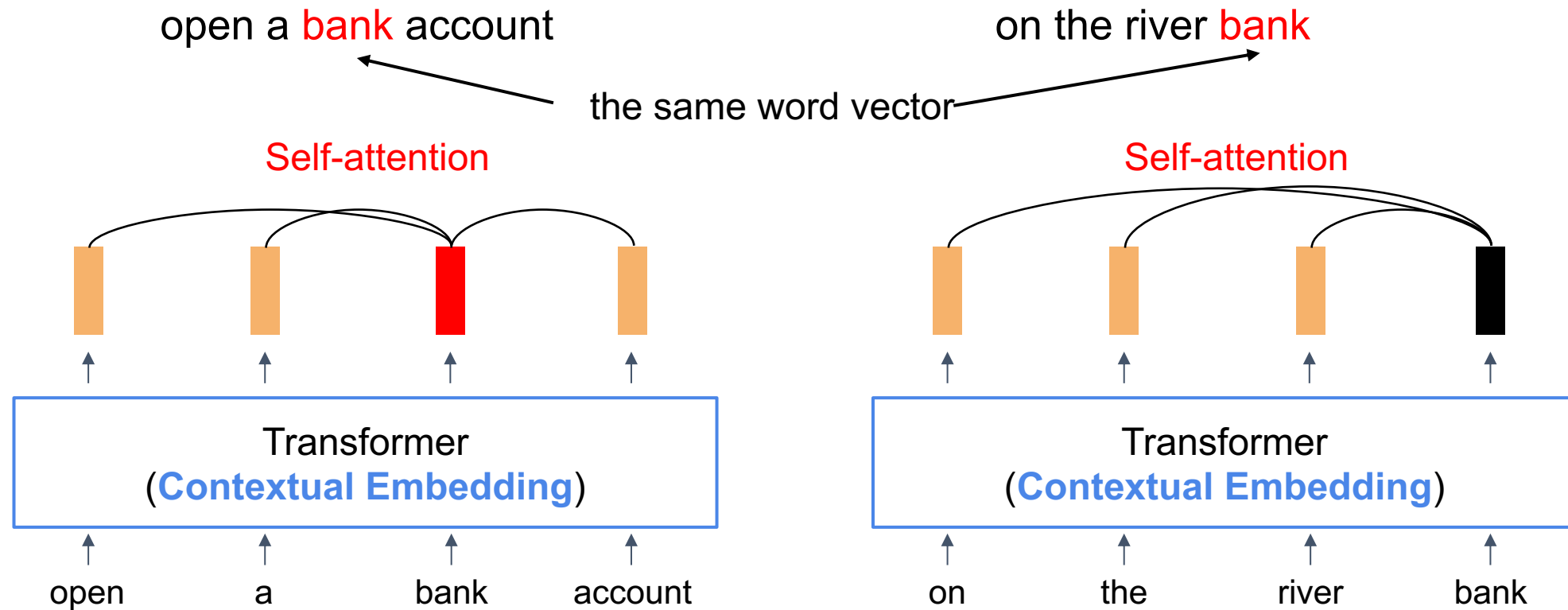
在目標資料上進行訓練，通常是監督式 (Supervised Training)，也就是需要有標註的資料才能進行模型訓練

Contextual Word Representations (RNN)



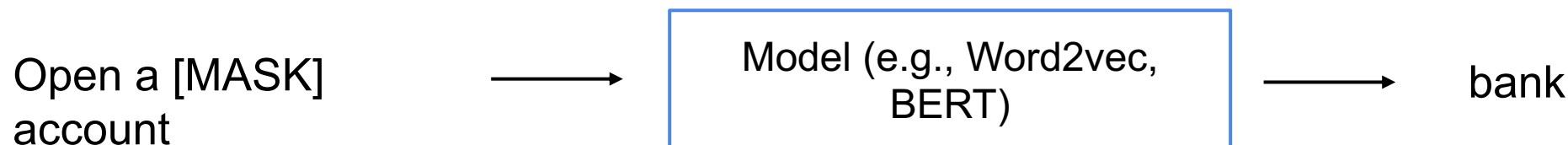
How about bi-directional?

Contextual Word Representations (Transformer)

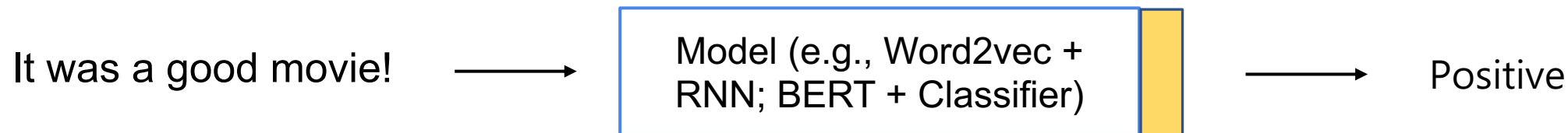


Model Training: Pre-training and Fine-Tuning

Step1: Pre-training (use large-scale corpora)



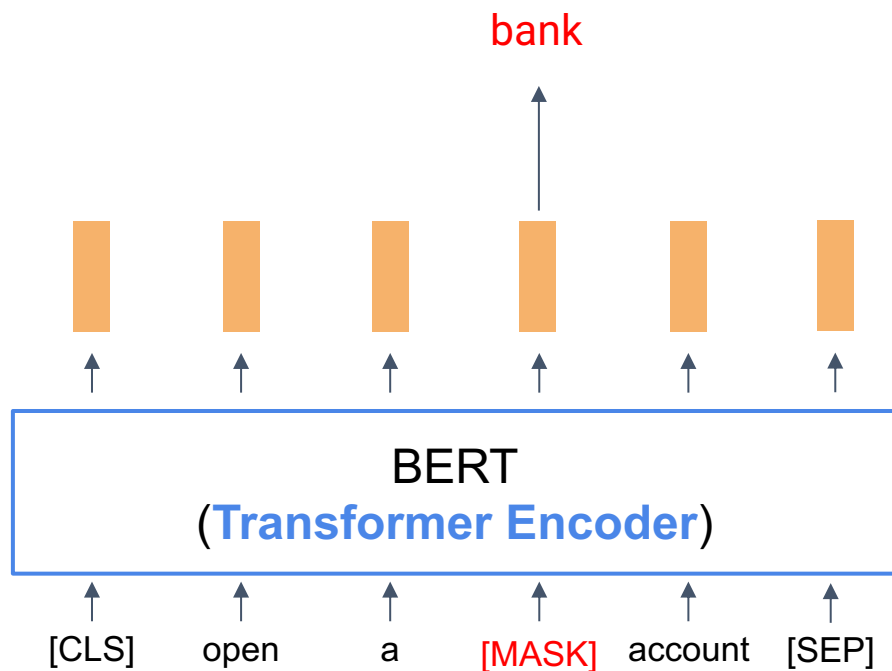
Step2: Fine-Tuning (use datasets from target tasks)



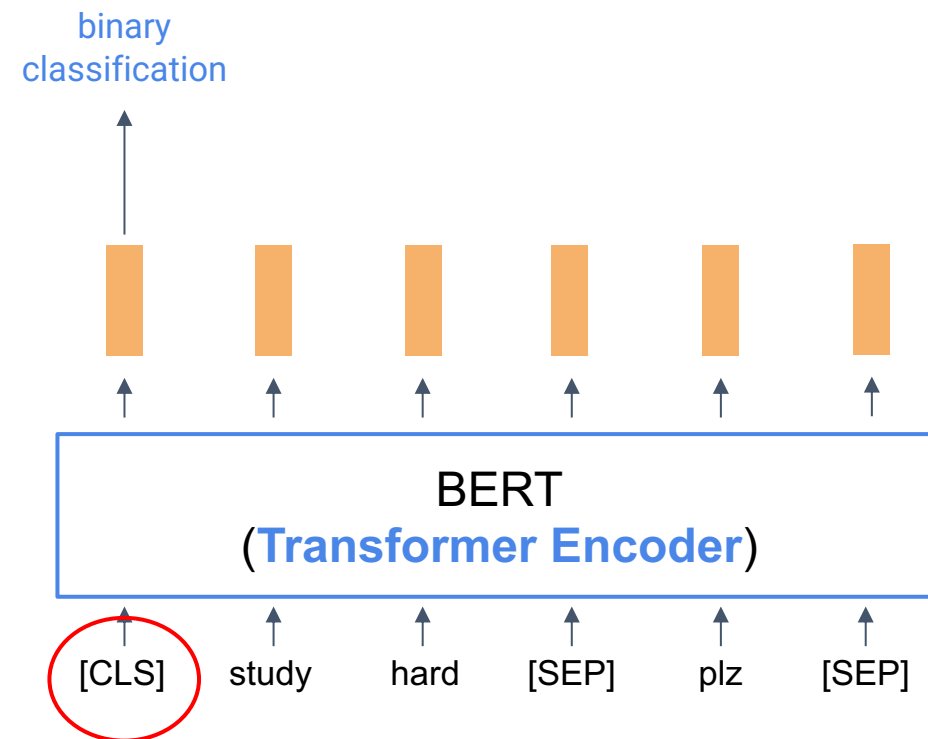
BERT (**B**idirectional Encoder Representations from Transformers)

- BERT 有兩種預訓練任務：

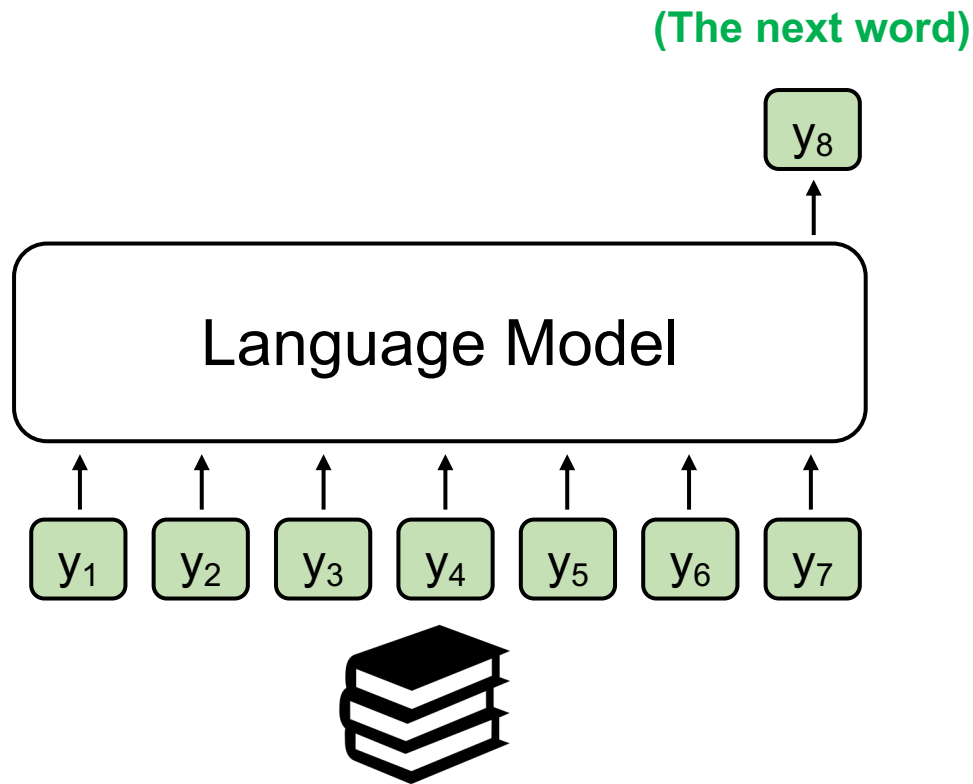
Masked Language Modelling



Next Sentence Prediction



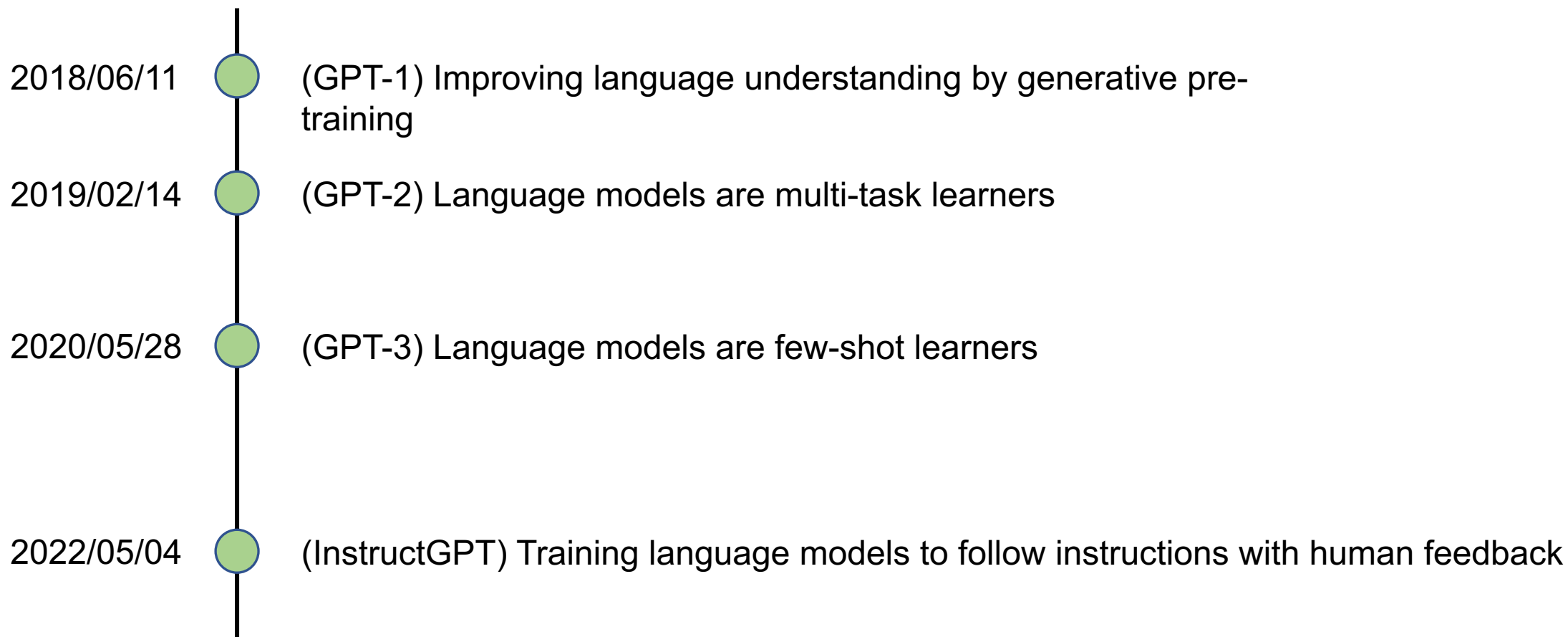
Generative Pre-Training (GPT)



$$P(y_t | y_1, y_2, \dots, y_{t-1})$$

- A model that assigns probabilities to upcoming words is called a **language model**.
- The task involving predictions of upcoming words is **language modeling**.

GPT 系列作品（有透露細節的）時間線



以前的 GPT 有口吃

InstructGPT: ChatGPT 的前身

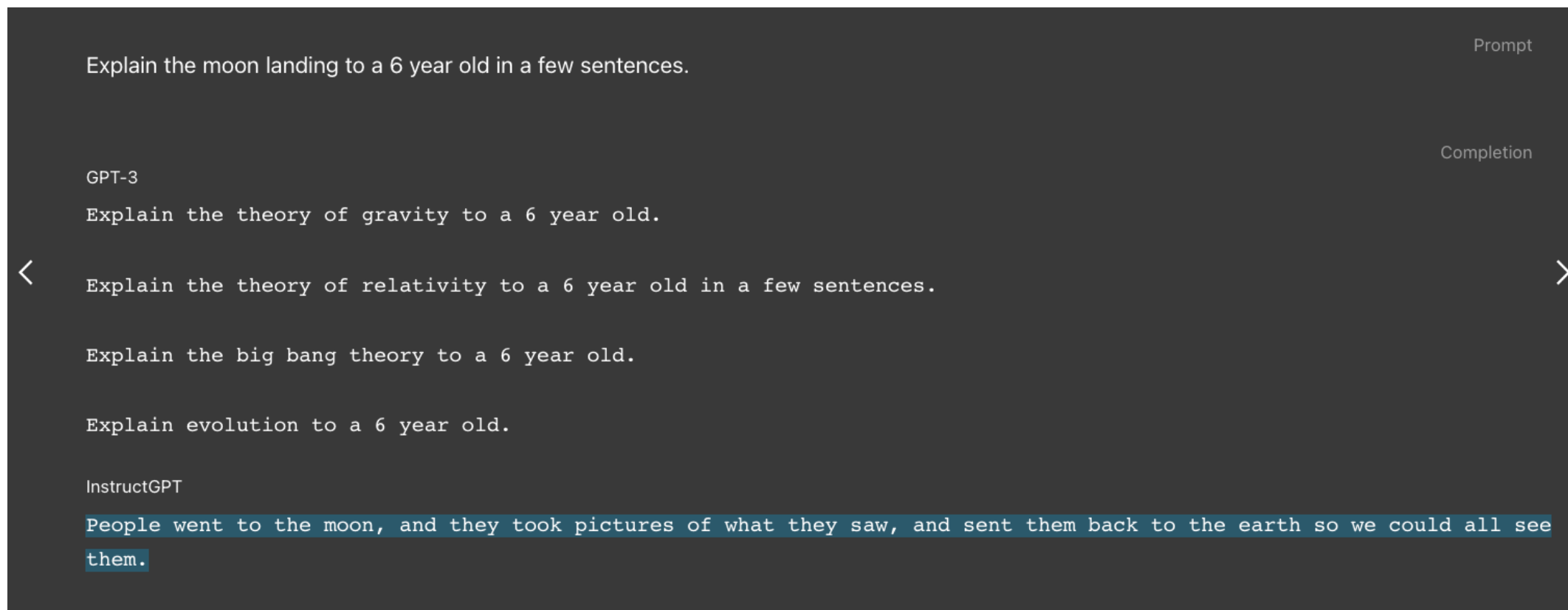
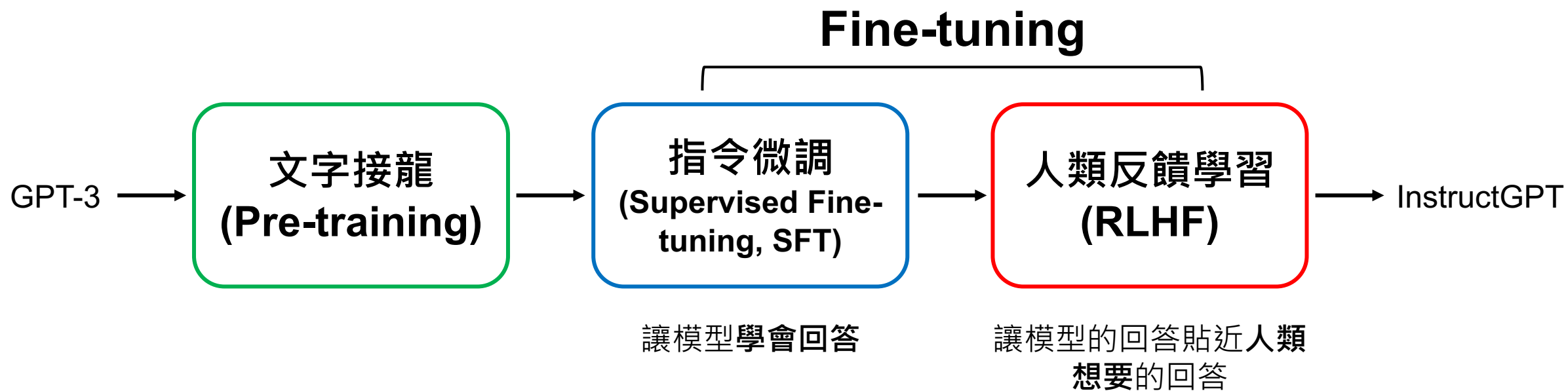


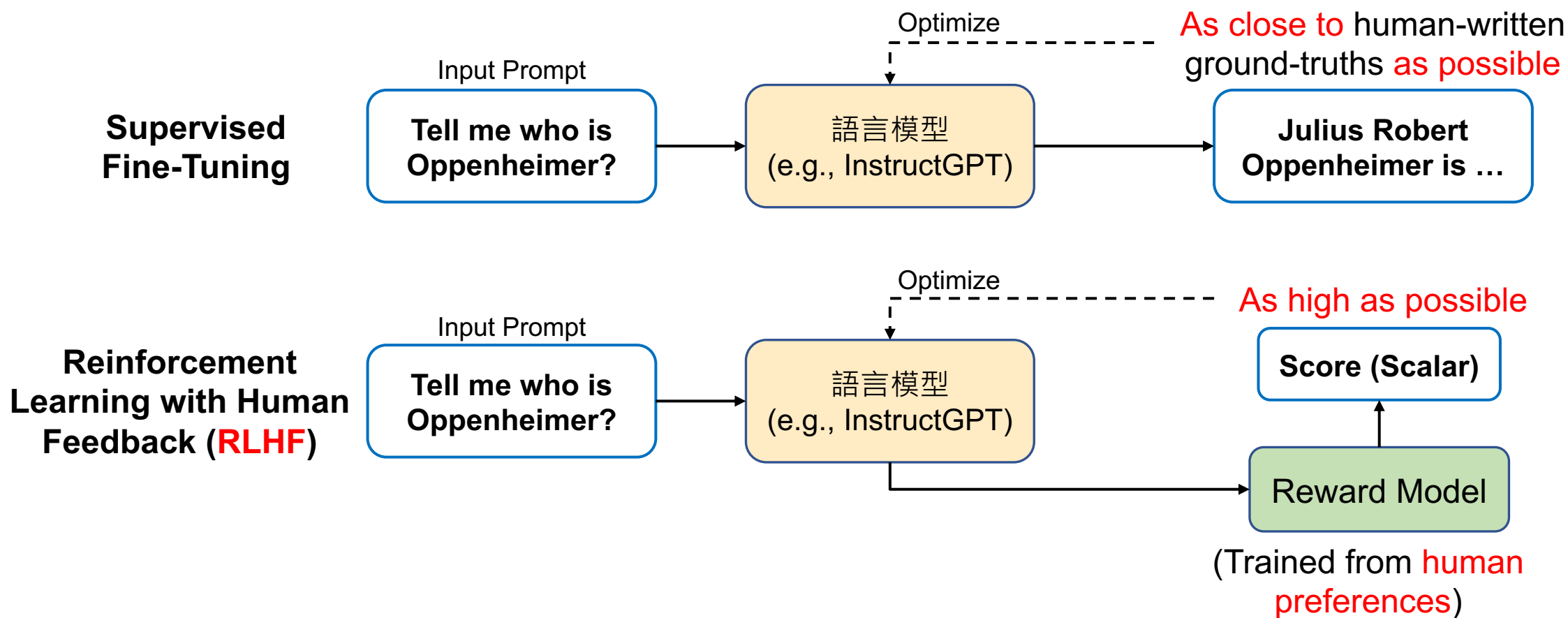
Figure source: <https://openai.com/index/instruction-following/>
Ouyang, Long, et al. "Training language models to follow instructions with human feedback." NeurIPS 2022.

The Pre-training then Fine-tuning Paradigm



RLHF: Reinforcement Learning with Human Feedback

RLHF 的進行過程



Main Differences between BERT and InstructGPT

Model size:

- BERT: 110 million
- InstructGPT: 175 billion (~BERT*1590 times)

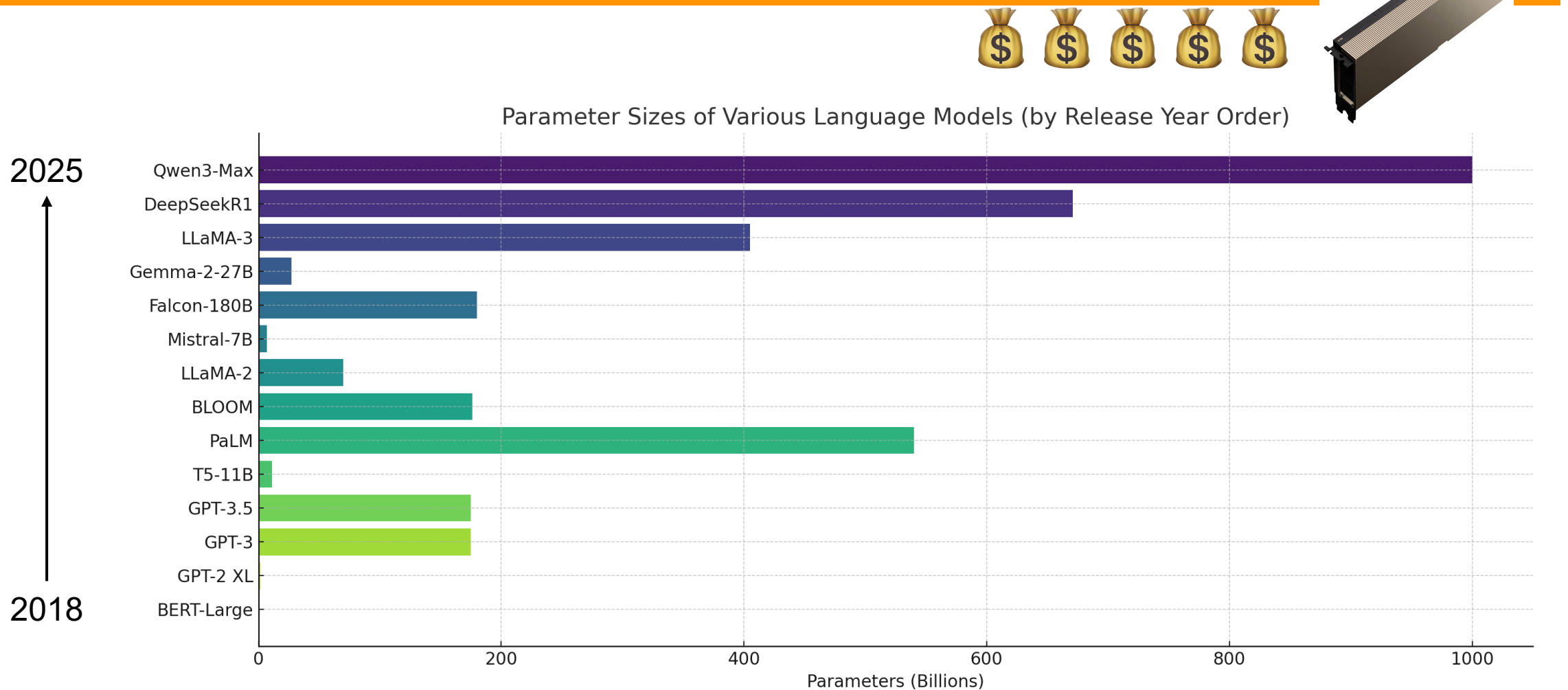
Training:

- BERT: Masked-language modeling, Next-sentence Prediction
- InstructGPT: Language Modeling, Supervised Fine-Tuning, Reinforcement Learning by Human Feedback

NLP Tasks:

- BERT: Natural Language Understanding (E.g., Classification)
- InstructGPT: Natural Language Generation (E.g., Machine Translation, Story Generation)

NLP Model Sizes (2018 – 2025)



The Revolution of ChatGPT

ChatGPT came out in November, 2022.

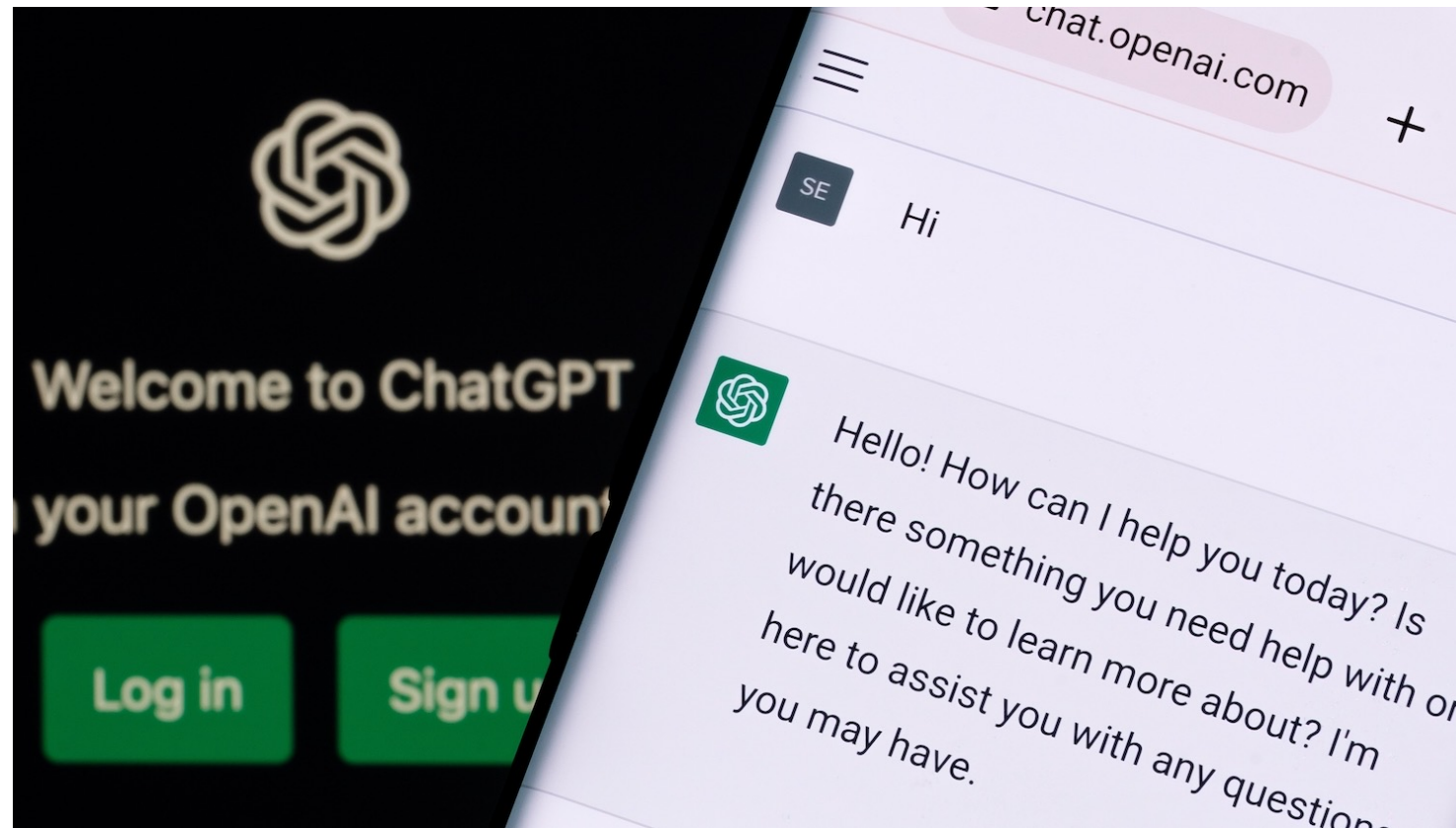
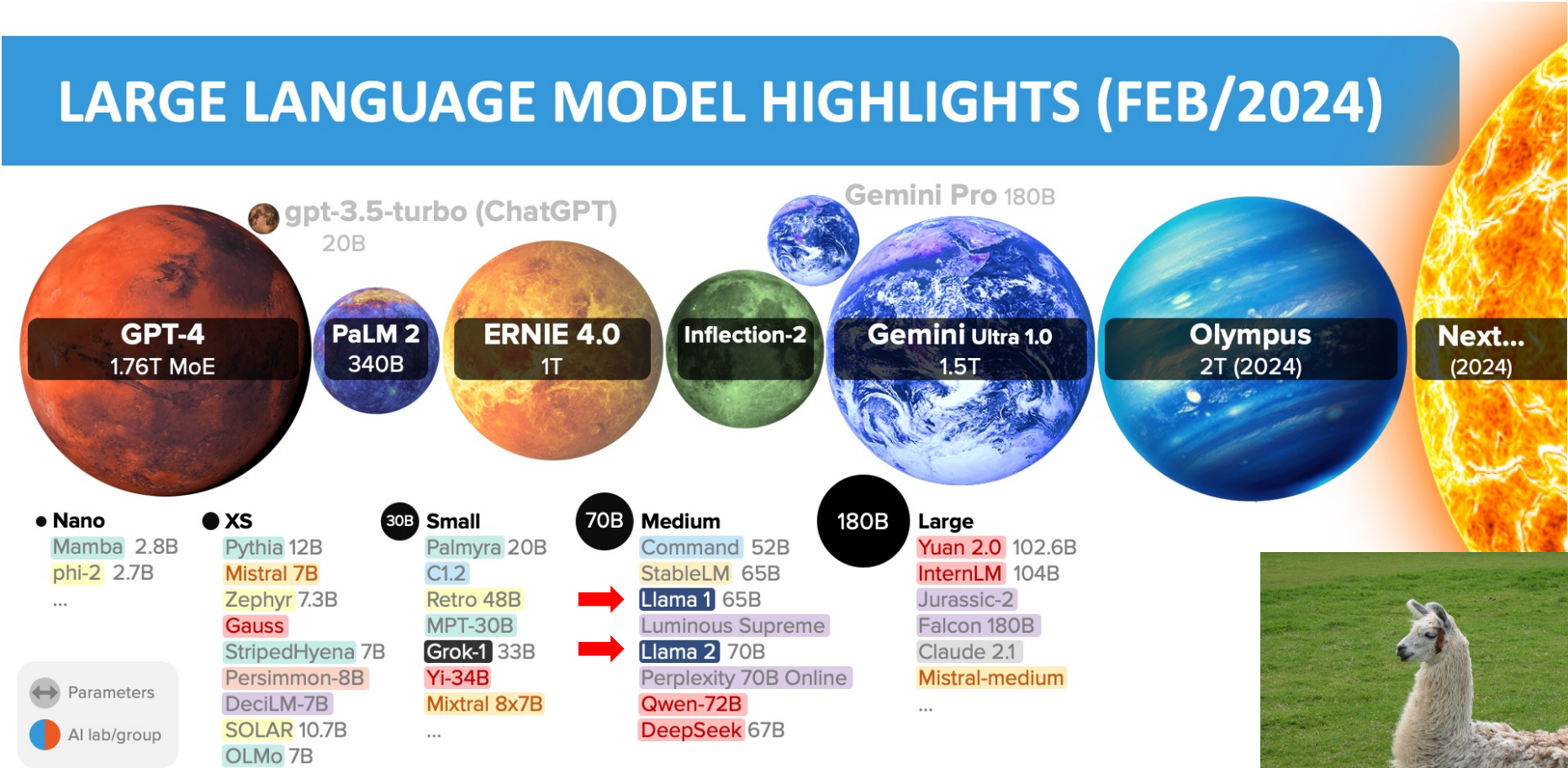


Figure source: <https://poole.ncsu.edu/thought-leadership/article/lets-chat-about-chatgpt/>
<https://openai.com/index/chatgpt/>

LLM Size Highlights



Sizes linear to scale. Selected highlights only. All models are available. All models are Chinchilla-aligned (20:1 tokens:parameters) <https://lilearchitect.ai/chinchilla/> All 200+ models: <https://lilearchitect.ai/models-table/> Alan D. Thompson, 2023-2024.

Source: Inside language models (from GPT-4 to PaLM) – Dr Alan D. Thompson – Life Architect



很多厲害的LLMs都需要上網且不公開...



Llama (7B - 65B) by Meta (2023)



- Llama1:
<https://arxiv.org/abs/2302.13971>
- Llama2:
<https://arxiv.org/abs/2307.09288>
- Llama3:
<https://arxiv.org/abs/2407.21783>

大型語言模型會思考？

ChatGPT 5.2 Thinking ▾

↑ 分享 ... 活動

14	State Space Models and Mamba	(HW4 due)
15	小組實作成果報告 (1)	
16	小組實作成果報告 (2)	

謝謝！接著能麻煩你為我的syllabus製作一份README嗎？我想要盡量帶點生動活潑的感覺，不然一般的GitHub repo太冷冰冰

我會把你這份 16 週課綱整理成一份「好讀、帶點活潑感」的 GitHub README：包含課程簡介、學習目標、每週主題表、作業/考試節奏、專案說明、資源與修課建議，並提供可直接貼到 repo 的 Markdown。

已思考 1m 17s >

下面是一份你可以直接貼到 repo 的 README.md 範本（走「生動但不油」路線 😊）。你只要把學期資訊、評分比重、聯絡方式等欄位補上就行。

Markdown

<!-- 建議：把封面圖放到 assets/banner.png，然後就能顯示在最上方 -->

想問就問

+ 思考 ▾

ChatGPT 可能會出錯。請查核重要資訊。

正在思考

- 處理圖片並轉錄表格內容
我應該寫一個讀取圖片的 README，並且內容應該在中文中呈現生動。需要從圖片中擷取排程表，手動轉錄第 1 至第 16 周的主題和注釋。打開圖片檢查，使用 Python 程式確認內容。

打開並檢視圖片

python 永遠顯示詳細資料 複製程式碼

```
from PIL import Image
img = Image.open('/mnt/data/e7ed1bb1-5
img.size
```

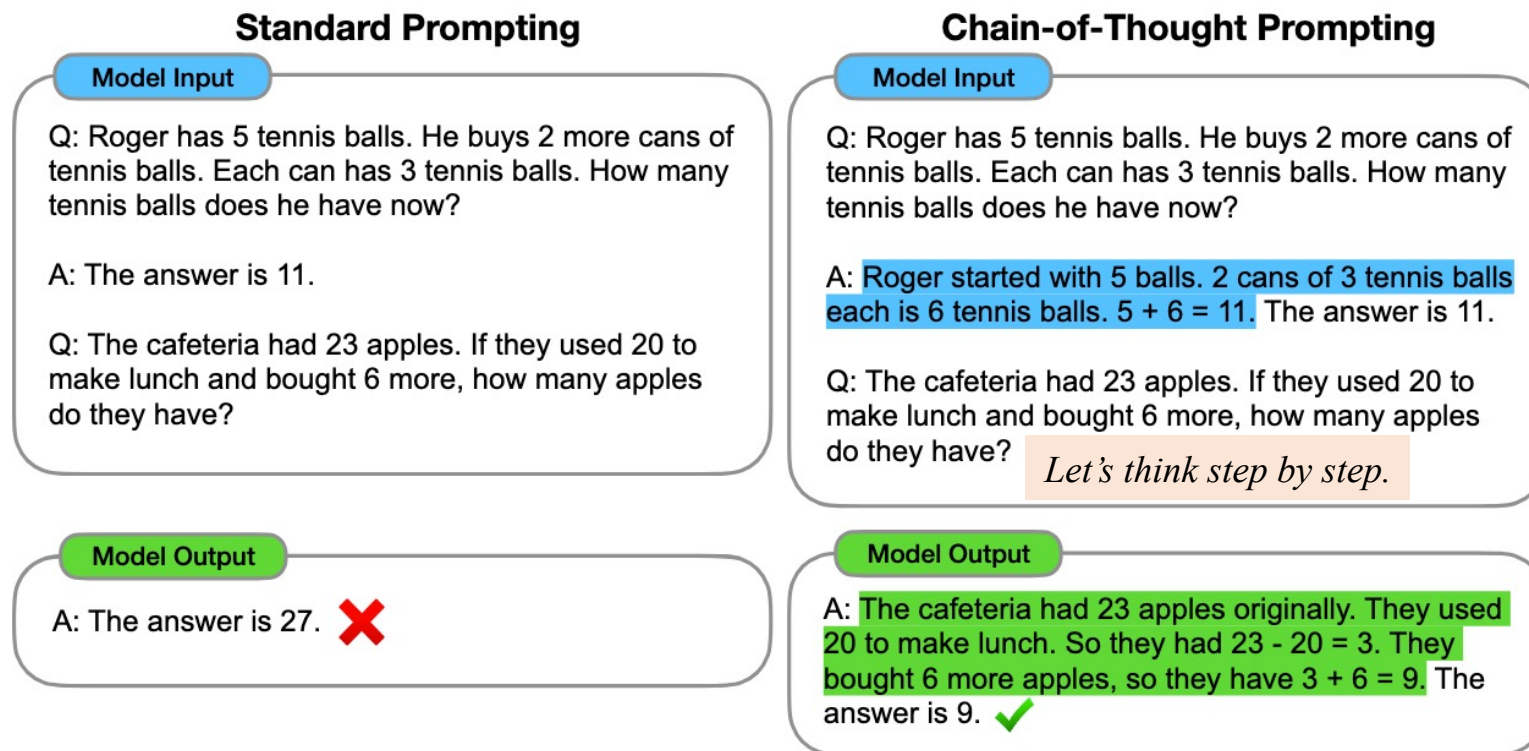
```
-----
FileNotFoundError                                Tra
Cell In[1], line 2
      1 from PIL import Image
----> 2 img = Image.open('/mnt/data/e7ed1bb1-
      3 img.size

File /opt/pyenv/lib/python3.11/site-packages
3511 if is_path(fp):
3512     filename = os.fspath(fp)
-> 3513     fp = builtins.open(filename, "rb"
3514     exclusive_fp = True
3515 else:

FileNotFoundError: [Errno 2] No such file or
```

- 檢查圖片路徑與上傳環境

Chain-of-Thought (CoT)



CoT:

<https://arxiv.org/abs/2201.11903>

Zero-shot-CoT:

<https://arxiv.org/abs/2205.11916>

Figure 1: Chain-of-thought prompting enables large language models to tackle complex arithmetic, commonsense, and symbolic reasoning tasks. Chain-of-thought reasoning processes are highlighted.

Training with Reasoning Tasks

The denominator of a fraction is 7 less than 3 times the numerator. If the fraction is equivalent to $2/5$, what is the numerator of the fraction? (Answer:)

Let's call the numerator x .

So the denominator is $3x-7$.

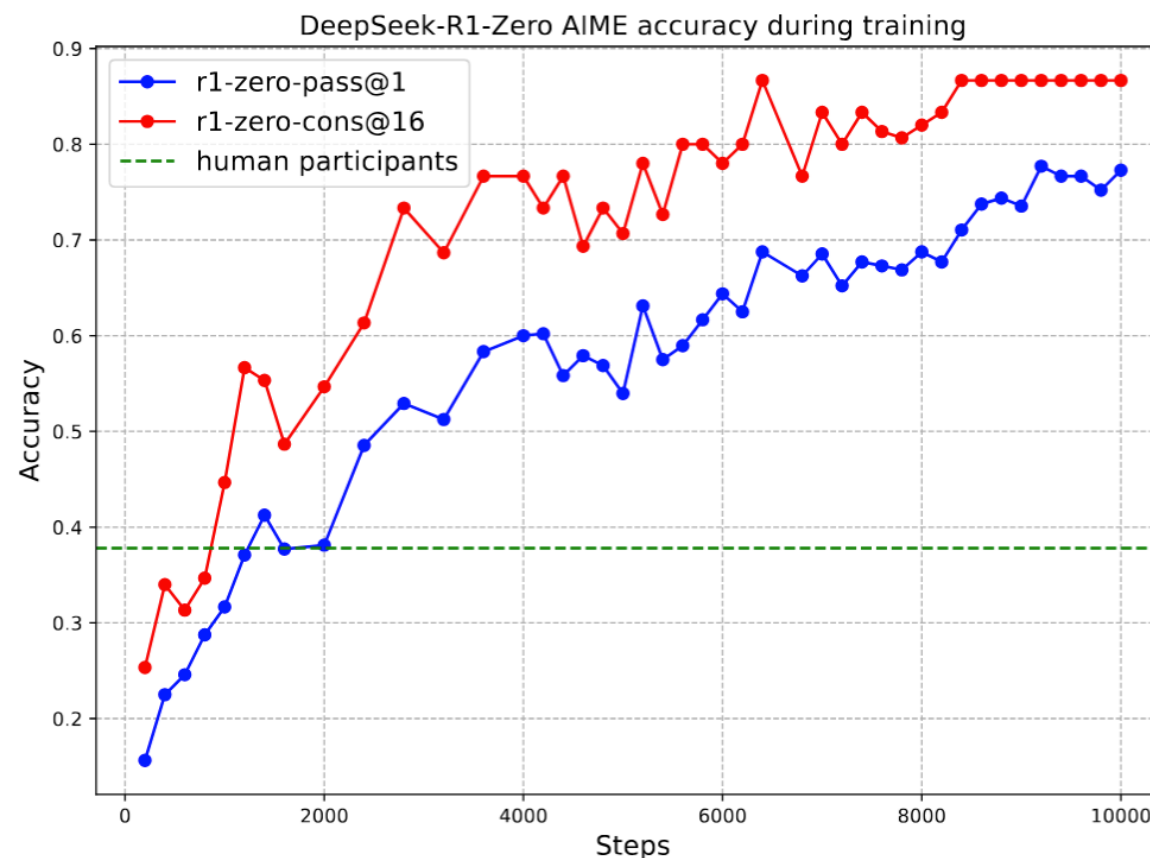
We know that $x/(3x-7) = 2/5$.

So $5x = 2(3x-7)$.

$5x = 6x - 14$.

So $x = 7$.

Let's Verify Step by Step:
<https://arxiv.org/abs/2305.20050>
DeepSeek-R1:
<https://arxiv.org/abs/2501.12948>



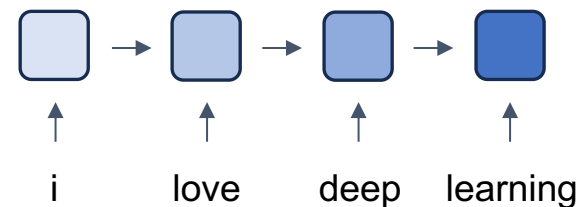
我們這學期要學什麼？

這門課的目標

- 能夠「理解」主流 NLP / LLM 方法在做什麼
 - 不是只會背名詞、不是只會看別人的部落格 / 重點整理
- 能夠運用現成工具（尤其是 🤗 Transformers 生態）把模型跑起來、改得動
- 知道如何進階 ➡ 評估、幻覺、安全性、RAG
- 如果你曾經覺得「LLM 很神，但我不知道它到底在神什麼」
—— 那這門課就是你的解謎本 📖🧩

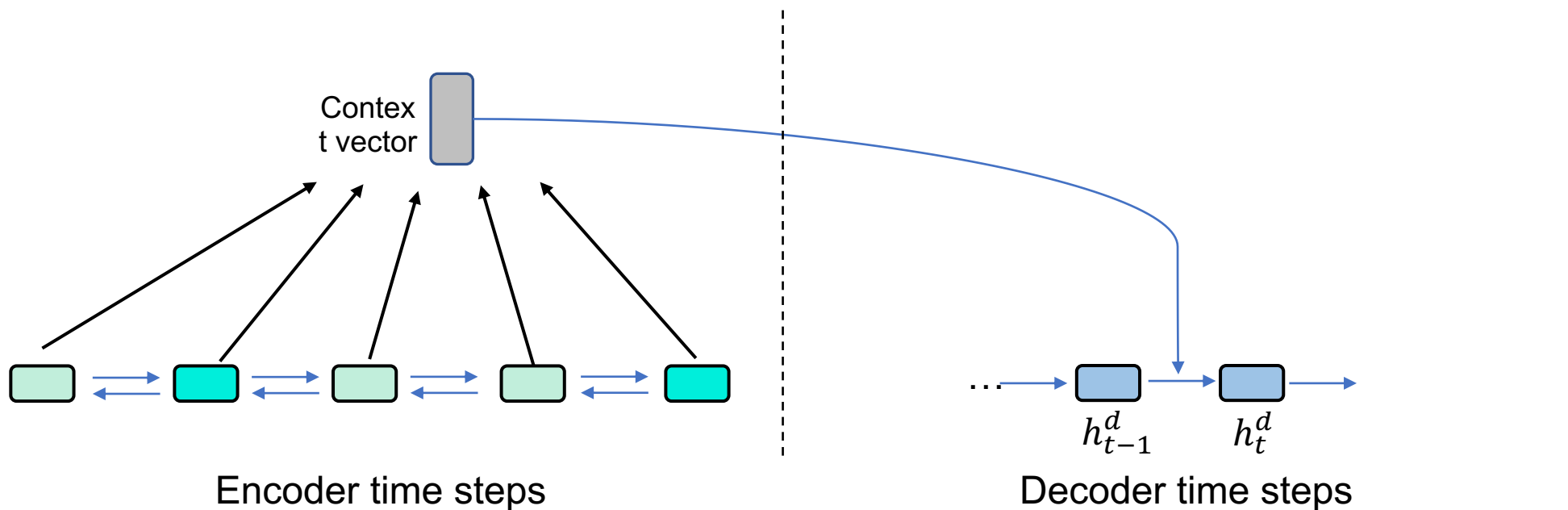
1. 自然語言處理基本功 (如何表達字詞)

- Week 1 – Week 3
 - 自然語言處理介紹
 - 文字向量基礎與詞嵌入模型
 - 遞迴神經網路與長短期記憶網路



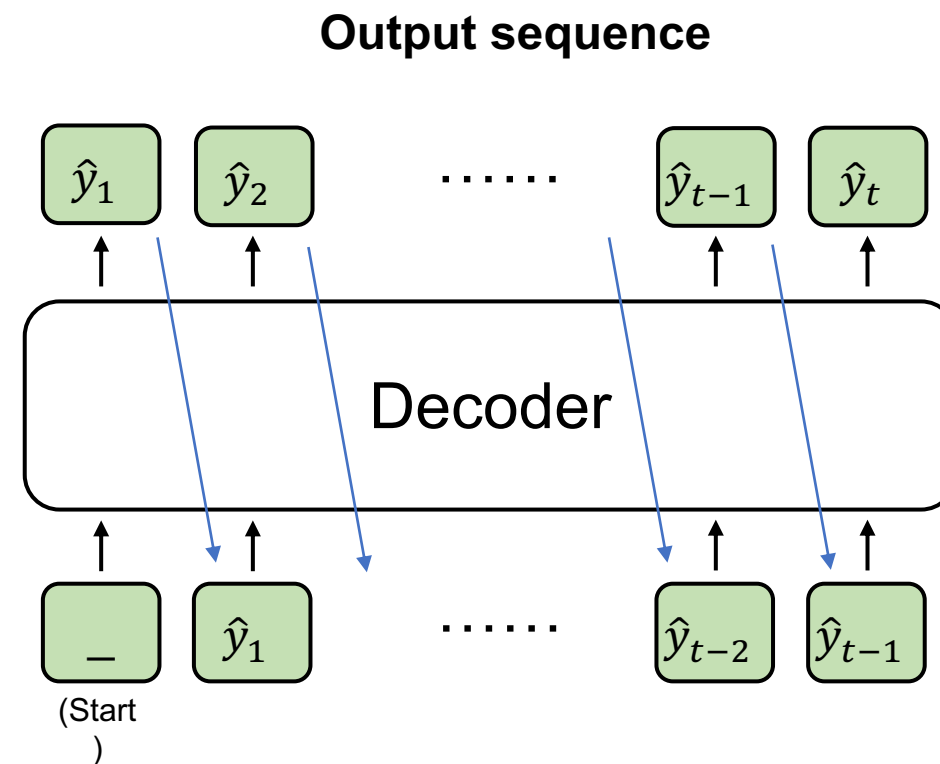
2. 現代自然語言處理基本功

- Week 4 – Week 5
 - 注意力機制以及 Sub-word 分詞法
 - Transformer 架構

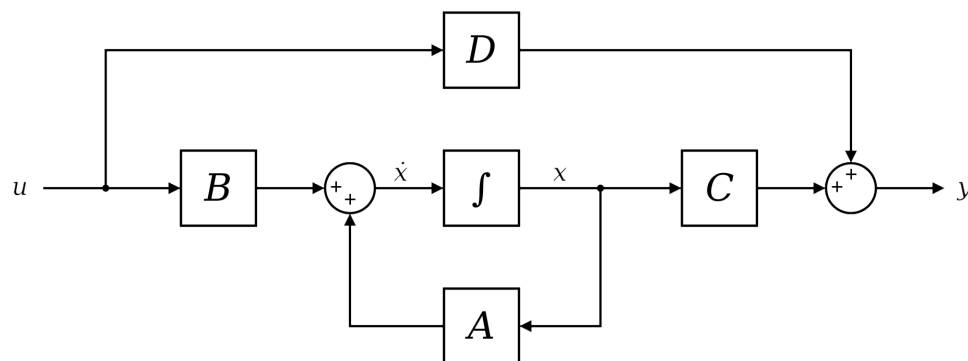


進階自然語言處理 (1)

- Week 6 – Week 11
 - BERT 🥰、GPT、文字生成策略
 - HuggingFace 函式庫實作教學
 - InstructGPT, RLHF, LLM
 - PPO, GRPO
 - Retrieval-augmented Generation



- Week 12 – Week 14
 - NLP Evaluations
 - Hallucinations? Safety?
 - 如何省力訓練大模型？ ➡ Test-time Compute
 - State Space Models and Mamba



➡ Figure source:
<https://huggingface.co/blog/lbourdois/get-on-the-ssm-train>

Common NLP tasks

**Text
Classification**

Sentiment Analysis

Natural Language
Inference (NLI)

**Text
Generation**

Machine
Translation

Summarization

Dialogue
Generation

**Sequence
Tagging**

Part-of-speech
tagging

Named Entity
Recognition (NER)

Word Segmentation

Text Matching

Semantic Textual
Similarity

Retrieval

我可以修這門課嗎？

- 建議有以下基礎：
 -  基礎 Deep Learning
 -  基礎 Python
 -  一點點機率/線代概念
- 願意花時間學習知識、學習除錯、學習驗證

Syllabus

Tutorial

Exam

Report

Week	Topic	Note
1	自然語言處理介紹以及本課程綱要	
2	文字向量基礎與詞嵌入模型	HW1
3	遞迴神經網路與長短期記憶網絡	
4	注意力機制與現代字詞分割演算法	
5	Transformer 模型架構介紹	HW2 (HW1 due)
6	BERT 與它的好夥伴：基於編碼器架構的 Transformer 模型	
7	清明連假	
8	GPT-1 到 GPT-3：基於解碼器架構的 Transformer 模型 文字生成策略	HW3 (HW2 due)
9	期中考	
10	HuggingFace 函式庫實作教學	
11	大型語言模型專題 1：指令微調與基於人類回饋的強化學習 & Retrieval Augmented Generation (RAG)	HW4 (HW3 due)
12	大型語言模型專題 2：如何評量模型的成效 / 幻覺 / 安全性？	
13	大型語言模型專題 3：如何訓練大模型？輕量化微調方法	
14	State Space Models and Mamba	(HW4 due)
15	小組實作成果報告 (1)	
16	小組實作成果報告 (2)	

PyTorch Tutorial

- PyTorch Basics:
 - <https://youtu.be/nJpkmWbteTU?si=Ay2EEiXIWHDL9D-K>
- PyTorch Modeling:
 - https://youtu.be/NEBJqya2IDs?si=-xX7dC_T3mDSAqjf
 - <https://youtu.be/41r4nmeGitk?si=veztilLno9PUZuex>

Scoring

- Assignments (4 times): 30%
- Quizzes: 1% x 10 times
- Mid-term Exam: 30%
- Term Project: 30%

About the Quizzes

- 不定時，代表出席分數，若多於10次即僅採計10次
- 與當週的上課內容有關
- Feel free to search the Internet

About the Assignments

- 與當週的上課內容有關
- 作業一定都會有範例程式碼 (Sample Code)

Final Project 是什麼？

- Final Project 是一種完整的訓練
- 你將自行從頭到尾做完一個題目，包含：
 - 資料前處理 (Data Pre-processing)
 - 模型訓練 (Model Training)
 - 模型評估 (Evaluations) --> 了解問題並改進模型
- 每組人數待修課同學總數訂定，原則上最多 4 人/組

Textbook

Speech and Language Processing (3rd ed. draft)

[Dan Jurafsky](#) and [James H. Martin](#)

Here's our January 12, 2025 release!

This release has no new chapters, but fixes typos and also adds new slides and updated old slides. Individual chapters and updated slides are below.

[Here is a single pdf of Jan 12, 2025 book!](#)

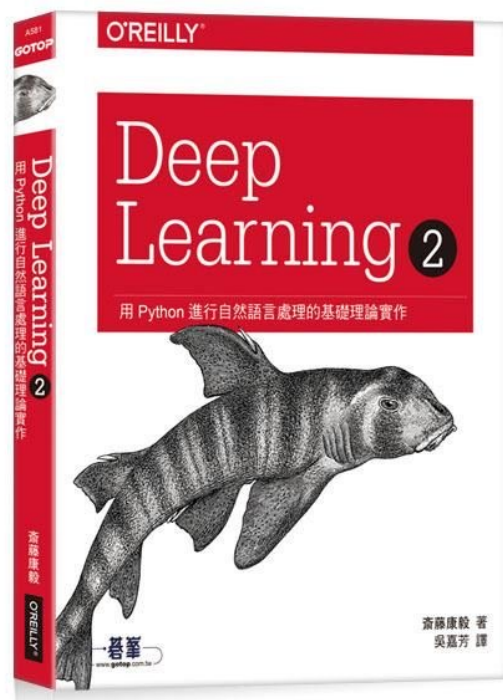
1. **Feel free to use the draft chapters and slides in your classes**, print it out, whatever, the resulting feedback we get from you makes the book better!
2. **Typos and comments** are very welcome (just email slp3edbugs@gmail.com and let us know the date on the draft)! (Don't bother reporting missing refs due to cross-chapter cross-reference problems in the individual chapter pdfs, those are fixed in the full book draft)
3. **Gratitude!** We've put up a [list here](#) of the amazing people who have sent so many fantastic suggestions and bug-fixes for improving the book. We are really grateful to all of you for your help, the book would not be possible without you!
4. **How to cite the book:**

Daniel Jurafsky and James H. Martin. 2025. Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition with Language Models, 3rd edition. Online manuscript released January 12, 2025. <https://web.stanford.edu/~jurafsky/slp3>.
5. A **bib entry** for the book is [here](#).
6. **When** will the book be finished? Don't ask.
7. If you need the previous Aug 2024 draft chapters, they are [here](#);

Speech and Language Processing (3rd ed. draft) Dan Jurafsky and James H. Martin.

Free Online book (Jan 2026 version):
<https://web.stanford.edu/~jurafsky/slp3/>

Reference book



Deep Learning 2 | 用Python進行自然語言處理的基礎理論實作，歐萊禮 (ISBN: 9789865020675)

GitHub of this book:

<https://github.com/oreilly-japan/deep-learning-from-scratch-2>

生成式AI融合教學

- 以組為單位
 - ~~ChatGPT Plus Claude Pro~~
 - Google Colab Pro 雲端運算付費服務
 - OpenAI /Gemini API 呼叫大型語言模型進行生成式服務

Contact Information

Instructor: 林英嘉

 yjlin@cgu.edu.tw

 Office Hour: 歡迎來信預約，一定沒問題

寄信前請在主旨加註記 [自然語言處理]

Thank you!

Instructor: 林英嘉

 yjlin@cgu.edu.tw

TA: 林宣辰

 b1128015@cgu.edu.tw